DP Biology – B3.2 Plant Systems

Transport in plants

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

Plant statements B3.2.7–10, SL and HL. Textbook pp. 303–310.

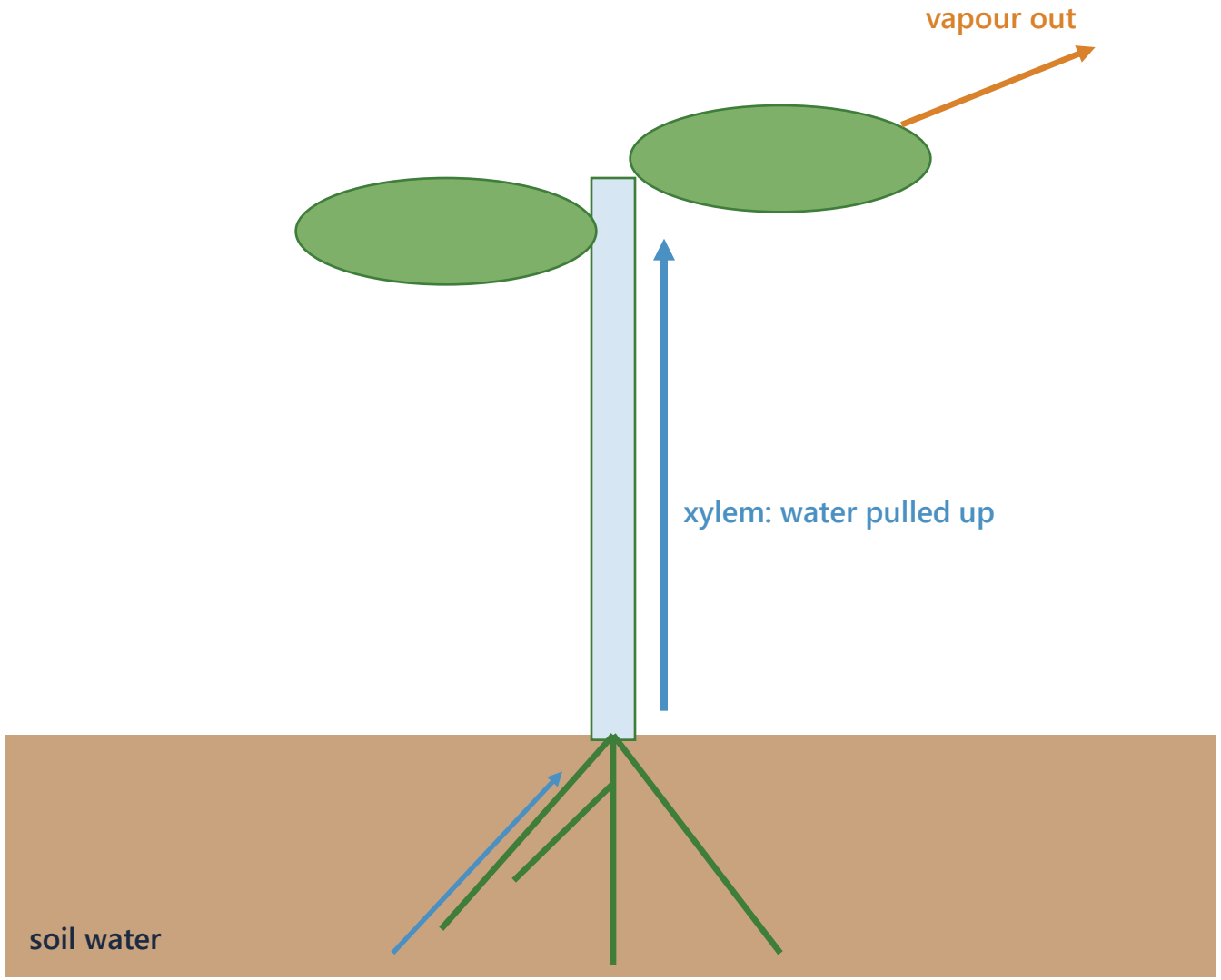


Image: part of a sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.7–8	Explain how transpiration draws water from roots to leaves, and how xylem is adapted to this
2	B3.2.9–10	Draw and annotate plan diagrams of dicot stems and roots from micrographs

From soil to air



Water **evaporates** from the leaves. This pulls more water up the **xylem** from the roots, which take it in from the soil.

This one-way flow is the **transpiration stream**. The plant spends no ATP on it: the Sun's energy drives evaporation.

Key idea

Water is pulled from above, not pushed from below.

Transpiration pulls water up

B3.2.7

Water evaporates from leaf cell walls → More water is drawn from the xylem through the walls → This creates tension in the xylem → Tension pulls the water column up the stem

Tension is a negative pressure: the water is being stretched. Because it is pulled from the top, the whole column moves up.

This explanation is called the **cohesion-tension** theory.

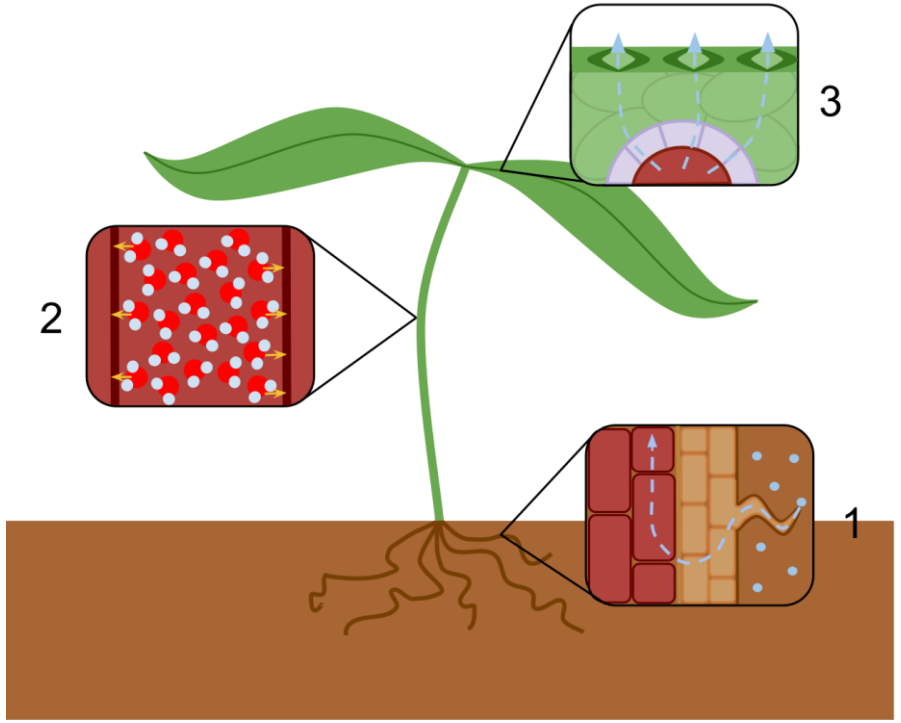
Common mistake

Evaporation in the leaves drives the flow. Roots do not pump water up.

Cohesion and adhesion

B3.2.7

Property	Role
Cohesion	Hydrogen bonds between water molecules hold the column together under tension
Adhesion	Hydrogen bonds between water and the walls draw water through cell walls and up the xylem



Water moving from roots to leaves. Laurel Jules, CC BY-SA 3.0.

Link

Both come from the polarity of water and its hydrogen bonds (A1.1).

Evidence that xylem is under tension

Shrinking trunks

Trunks get slightly narrower at midday, when transpiration is fastest.

Air drawn in

Cut a xylem vessel and air is sucked in; water does not spurt out.

Snapping columns

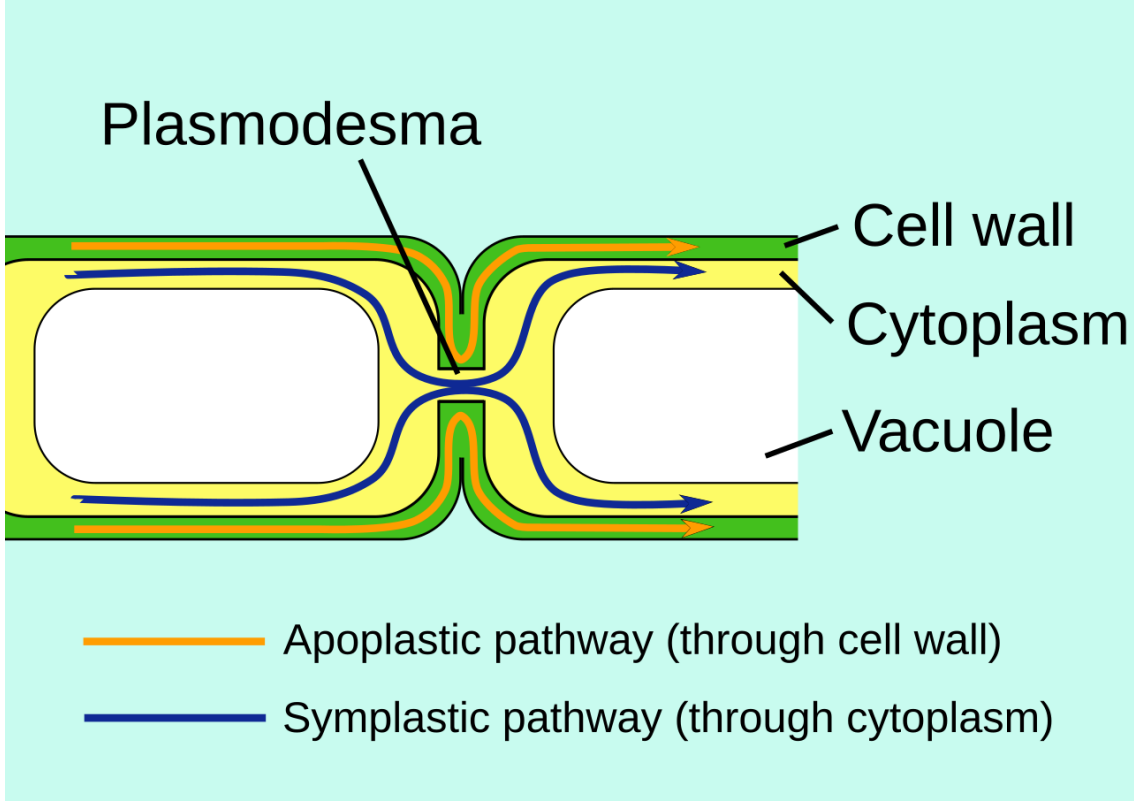
In drought, water columns break with clicks that sensitive microphones can hear.

Think

Predict how the trunk diameter changes during the night, and explain why.

Getting water into the root

B3.2.7



Apoplast and symplast routes across a root. Jackacon, Smartse, public domain.

Root hairs give a huge surface. Root cells take up mineral ions actively, and water follows by **osmosis**.

Water crosses the root through cell walls (**apoplast**) and through cytoplasm (**symplast**) to reach the xylem.

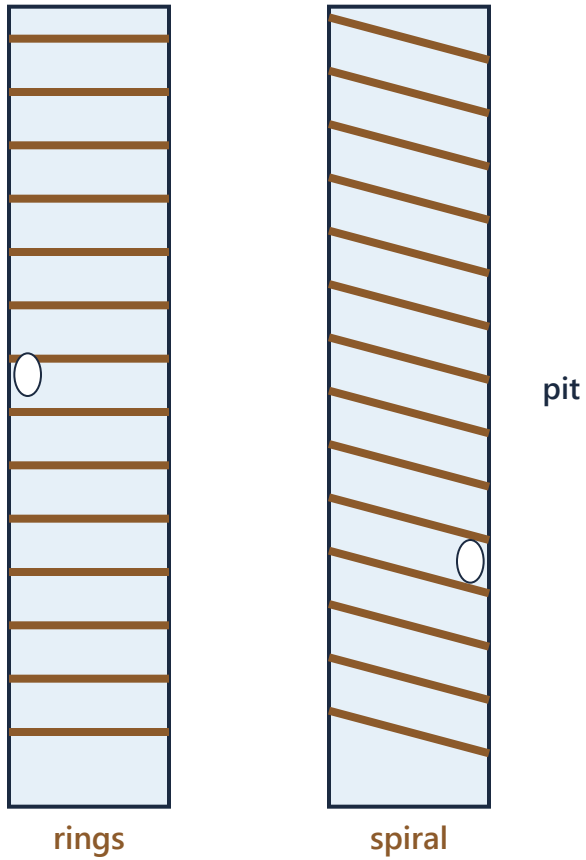
Beyond the syllabus

These routes complete the story; the statement itself focuses on tension and cohesion.

Xylem vessels are built for the job

B3.2.8

Feature	How it helps
No cell contents	Flow is unimpeded
No end walls	A continuous tube
Walls thickened with lignin	Resist collapse under tension
Pits in the walls	Water moves sideways



Key idea

Mature xylem is dead: hollow, reinforced tubes.

Working with data: sap flow over a day

B3.2.7

Speed of sap in the xylem of a tree trunk on a sunny day. These data are invented for practice; the pattern matches real measurements.

Time of day	Sap speed / cm h^{-1}
04:00	2
08:00	12
12:00	38
16:00	30
20:00	8

1. Describe the pattern.
2. Explain the peak.
3. Calculate how far sap moves in 2 hours at noon.
4. Suggest why sap still moves slowly at night.

Transport in animals and plants

B3.2.7

	Mammal	Flowering plant
Vessels	Arteries, capillaries, veins	Xylem vessels
Living?	Walls of living cells	Dead, hollow tubes
Driven by	Heart: positive pressure	Evaporation: tension
Energy from	ATP in heart muscle	The Sun
Backflow stopped by	Valves	Cohesion of the column

Link
This answers the second guiding question. The mammal side is B3.2.1–6.

Practice: predict and explain

B3.2.7–8

Leaves coated

Petroleum jelly is spread over the lower surface of every leaf.

Hot, windy afternoon

Wind and heat increase on a dry day.

Weak walls

A mutant plant makes much less lignin.

For each one

Predict the effect on water movement up the xylem and explain it, using evaporation, tension and cohesion.

Check-in: class 1

B3.2.7–8

1. Explain how transpiration pulls water up the xylem.
2. Distinguish between cohesion and adhesion.
3. Give one piece of evidence that xylem is under tension.
4. Give three ways xylem vessels are adapted.
5. State why plants use no ATP to move water up the stem.

How to use this

Answer without your notes, then check against slides 3–11.

Tissues in a dicot stem

B3.2.9



Sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

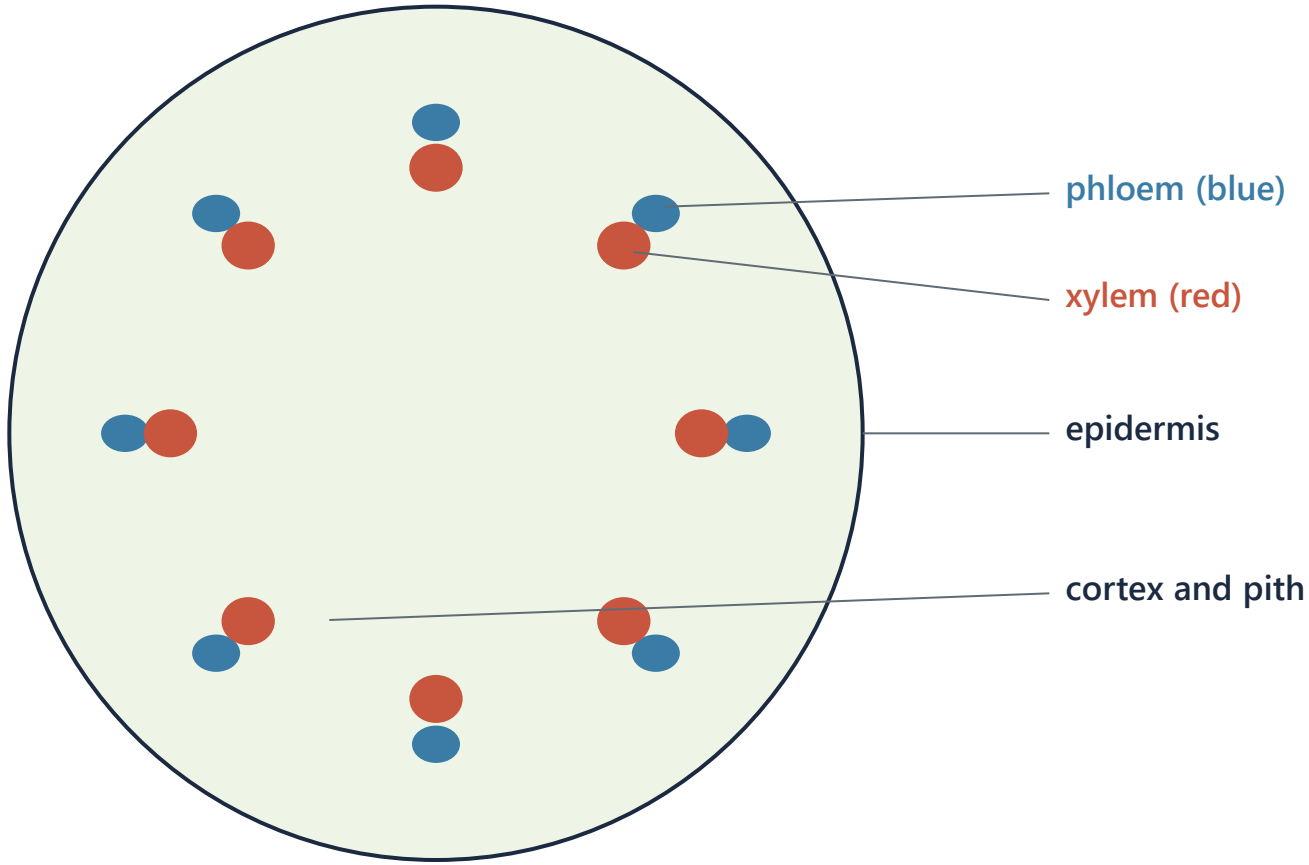
Tissue	Main function
Epidermis	Protects; reduces water loss
Cortex	Between epidermis and bundles; support
Vascular bundles	Ring near the outside; transport
Xylem	Inner part of each bundle: water, ions
Phloem	Outer part of each bundle: sugars

Remember

Stem: bundles in a ring near the edge, phloem outside the xylem.

Plan diagram: dicot stem

B3.2.9



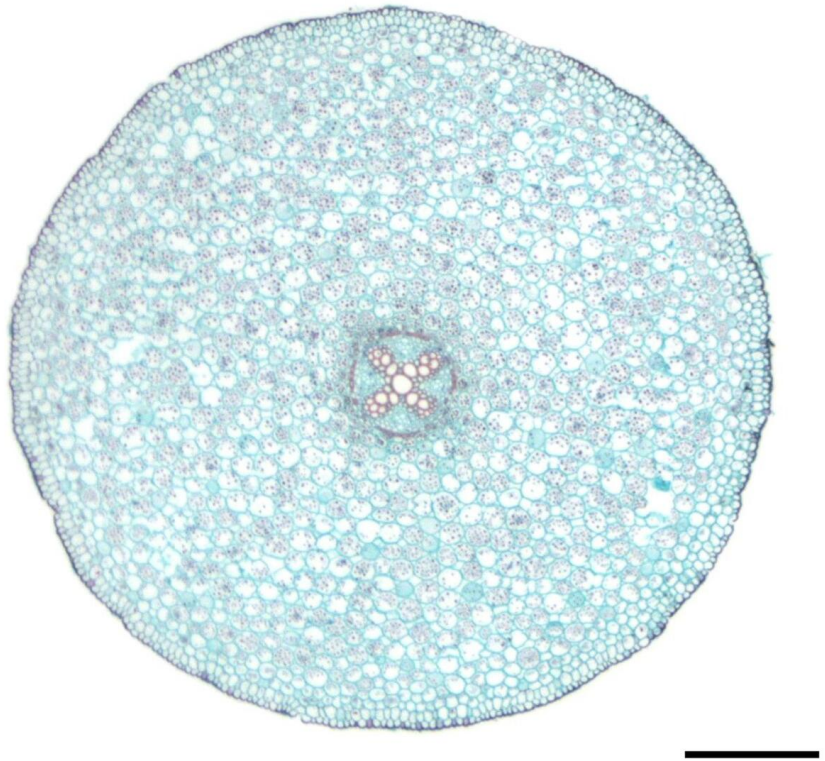
Drawn plan diagram, not to scale. In a real plan, draw outlines only, without colour.

Eight **vascular bundles** in a ring.

In each one, **xylem** faces the centre and **phloem** faces outwards.

Tissues in a dicot root

B3.2.10



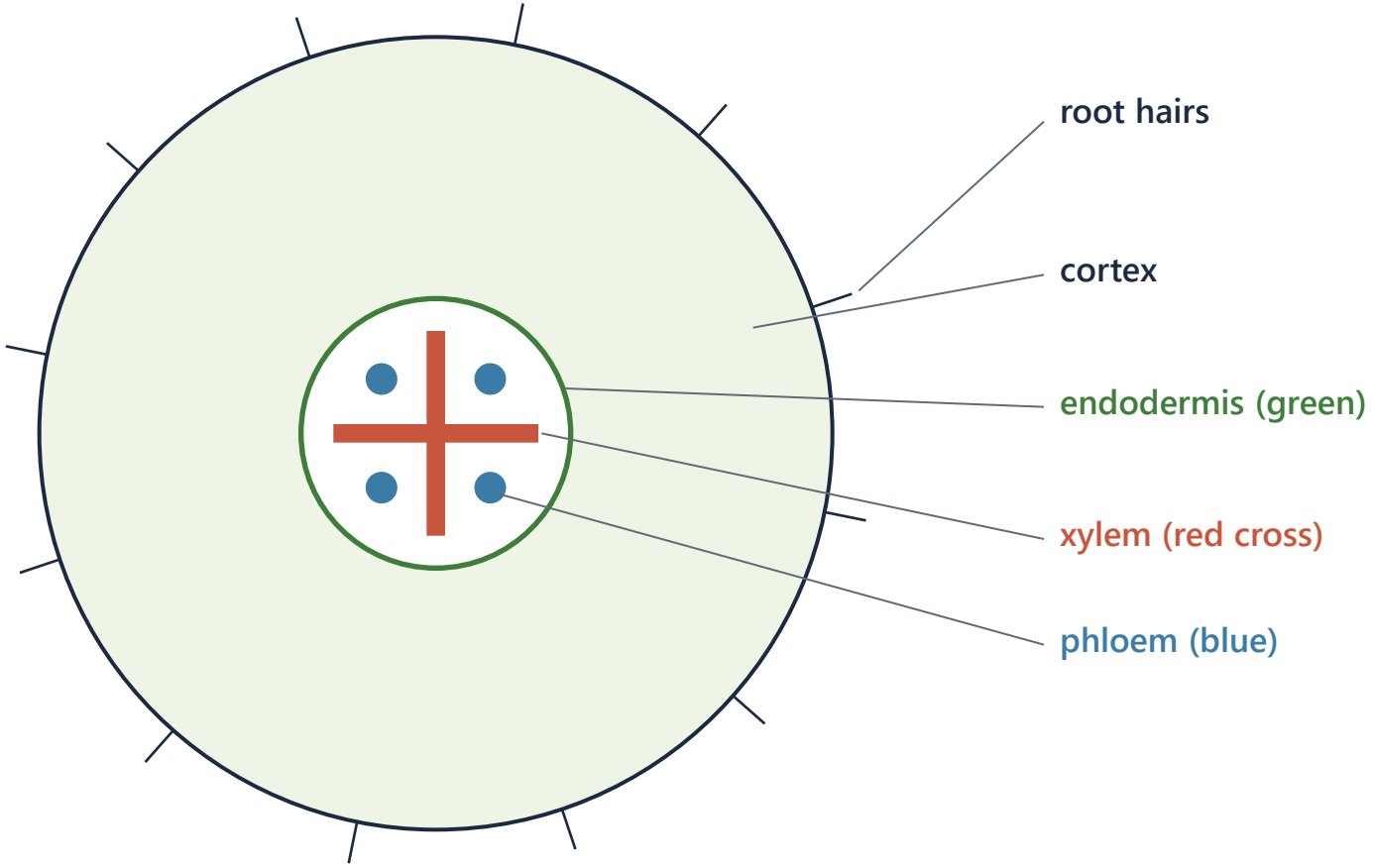
Buttercup (Ranunculus) root, stained section. Jon Houseman, CC BY-SA 4.0.

Tissue	Position and function
Epidermis	Outer layer; root hairs absorb water
Cortex	Wide region; stores food
Endodermis	Ring around the vascular tissue
Xylem	Star or cross at the very centre
Phloem	Between the arms of the xylem

Remember

Root: vascular tissue in the centre, xylem as a star.

Plan diagram: dicot root



Drawn plan diagram, not to scale. In a real plan, draw outlines only, without colour.

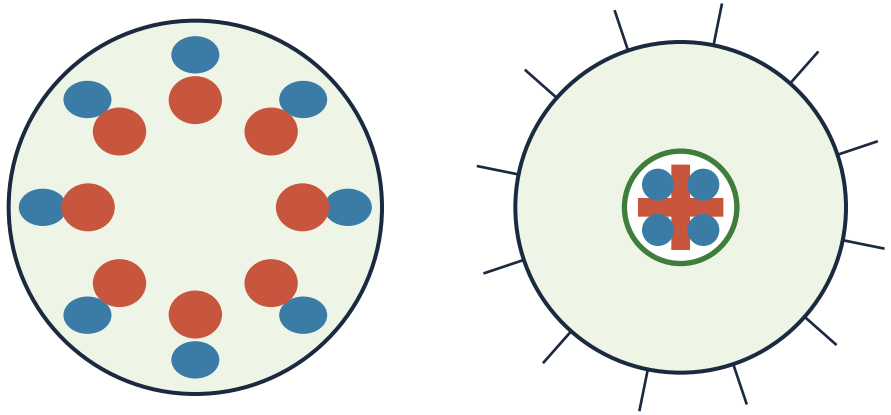
All the vascular tissue is in a **central core**.

Phloem sits between the arms of the **xylem**.

Drawing plan diagrams

B3.2.9–10

1. Outlines of tissues, not cells
2. Clear, continuous pencil lines
3. No shading
4. Keep the proportions
5. Label every region and add functions
6. Add a scale bar or magnification



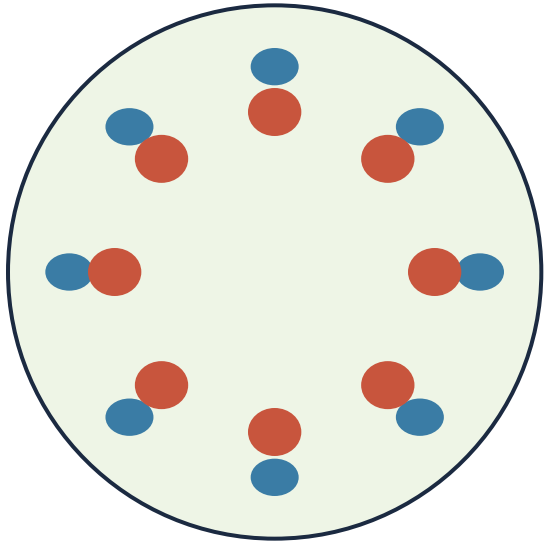
stem root

Activity

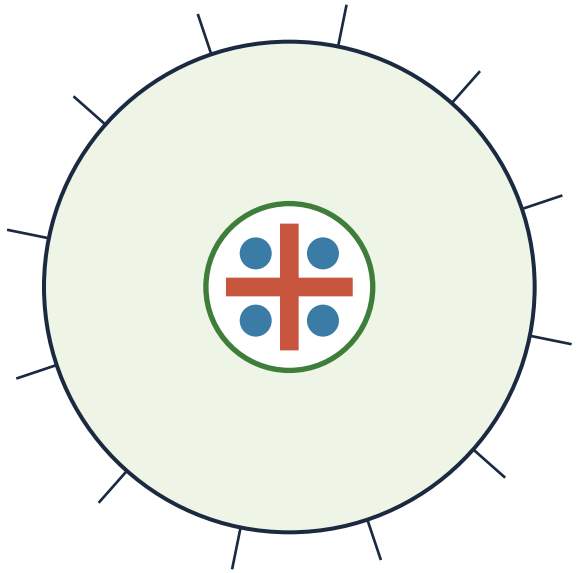
Draw plan diagrams of a dicot stem and root from micrographs, and annotate each tissue with its function.

Practice: stem or root?

B3.2.9–10



Section P



Section Q

1. Identify P and Q as stem or root.
2. Give two pieces of evidence for each.
3. Where is the xylem in each?
4. Explain why water enters Q at its edge.

How to use this

Answer from the drawings alone, then check against slides 13–16.

Practice: scale and magnification

B3.2.9–10

Magnification = size of drawing / actual size. Use the same units for both.

Measurement	Value
Real diameter of a root	2.0 mm
Diameter of the plan diagram	100 mm
Real diameter of a stem	5.0 mm
Stem drawing magnification	x 30

1. Calculate the magnification of the root drawing.
2. Calculate the diameter of the stem drawing.
3. A scale bar of 1 mm on the root drawing: how long should it be?

Check-in: class 2

B3.2.9–10

1. State where xylem and phloem are found in a dicot stem.
2. State where xylem and phloem are found in a dicot root.
3. Name the layer that surrounds the vascular tissue in a root.
4. List three rules for drawing a plan diagram.
5. A root 1.5 mm wide is drawn 75 mm wide. Calculate the magnification.

How to use this

Answer without your notes, then check against slides 13–19.

Common mistakes to avoid

Not this	But this
Water is pumped up the stem by the roots.	It is mainly pulled up by transpiration.
Xylem vessels are living cells.	Mature xylem vessels are dead, empty tubes.
Water enters roots by active transport.	Ions are taken up actively; water follows by osmosis.
Xylem is on the outside of a stem bundle.	Phloem is outside; xylem is inside.

Summary: transport in plants

B3.2.7–10

	Key idea
B3.2.7	Evaporation creates tension; cohesion and adhesion hold the column
B3.2.8	Xylem: dead, hollow, no end walls, lignified, pitted
B3.2.9	Stem: bundles in a ring, phloem outside xylem
B3.2.10	Root: central xylem star, phloem between the arms

Check that you can...

1. **Explain** how transpiration moves water up the stem
2. **Outline** the roles of cohesion and adhesion
3. **Explain** how xylem vessels are adapted
4. **Draw** plan diagrams of a dicot stem and root
5. **Calculate** magnification of a drawing
6. **Compare** transport in plants and animals

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Put in order: (A) tension pulls water up the xylem. (B) water evaporates from leaf cell walls. (C) water is drawn out of the xylem by capillary action.

2. Why do xylem vessels need lignin in their walls?

3. State one difference between where xylem is found in a dicot stem and in a dicot root.

Image credits

Helianthus stem.jpg – Jon Houseman – CC BY-SA 4.0

Transpiration Overview.svg – Laurel Jules – CC BY-SA 3.0

Apoplast and symplast pathways.svg – Jackacon, vectorised by Smartse – Public domain

Ranunculus root.jpg – Jon Houseman – CC BY-SA 4.0

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